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# Health Quality and Cost of Living in Asian Cities

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**Abstract**

Health is now a global concern. However, only a few studies are conducted on how the cost of living affects the quality of health care. This study aimed to determine the relationships between the cost of living and quality of healthcare. It is a quantitative method that utilizes data mining technique in the data collection. Numbeo database was used to determine the cost of living and healthcare indices of 36 Asian cities. Statistical treatment was employed using statistical software to generate patterns and to determine the relationships of these indices through regression analysis. The scatterplot revealed different patterns of the high cost of living (X) and health care index (Y), namely: 1) simple linear model (inverse relation), 2) simple linear model (directly proportional) and 3) elliptical pattern. Findings revealed that there are cities in Asia which allocate more in health care spending than other personal needs. In general, people living in Asia do not necessarily avail of health care programs and services regardless of their cost of living. Such relationship, therefore, differs from one city to another. Nevertheless, the price of living index and health care index relationships infers that there are cities in Asia that tend to increase its health care index when the cost of living increases. These people are health conscious and allocate a higher percentage of their expenses to health and wellness. There are also cities that have minimal budget allocation for health care from their GDP which requires their people to apportion more personal funds for health.

**Keywords:** Cost of living; health care; health care cost; health expenditure; health care quality

## Introduction

Health is a fundamental human right for the body and mind to function accurately. In fact, the World Health Organization (WHO) envisioned the achievement of health's highest quality. This health right and the similar quality of health care are crucial, and reliant upon, the realization of basic human needs as food, clothing, housing, work, education among others.

These basic human needs constitute the cost of living (COL) according to Numbeo, the world's leading database of user's contributing data about cities and countries. These necessities may also include groceries, restaurants, transportations, and utilities such as health care. These COL data are in the form of an index showing differences in living costs between cities. Higher COL Index suggests that the city is more expensive to live. Thus, health care quality in the Asian cities may be affected by the high or low cost of living.

On the other hand, Health care Index is an estimation of the overall quality of the health care systems, health care professionals, equipment, and health professionals, among others. This index tracks the movements in the health care expenditures, medical expenses, insurance coverage, health industry employment, international comparisons, and health care education (Cook, 2015).

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Studies conducted on health care mostly focuses on the correlation between health care quality and health care cost. The recent Supreme Court of Canada found out that quality of health care combined with government policy affects costs of health care (Mitton et al., 2006).

Another study by Burke and Ryan (2014), disclosed that the high cost of health care poorly ranks on other indicators such as terms of life expectancy. Further, low health care value was found to be mainly attributed to the regional variations in spending patterns and were not cost effective. The paper also summarized that efforts must be made to reduce the unwanted difference in care.

In developed Asian cities, health care spending that equates to health care cost, is higher in developing Asian cities. As health care expenditure or cost factors were examined, influence was pointed out to be dependent on the demand, supply, and other factors. There is an insignificant relationship between price and health care's cost, and positive and a significant connection between gross domestic product (GDP) and health expenditure. The increased health expenditures initiated the urban population density in other countries. Similarly, out-pocket payments have contributed to higher health care expenditure. The study summarized that health care expenditure or cost that continuously increases is brought about by changes in all factors over the period (Kraipornsak, 2017).

The studies cited mostly focus on the cost of health care and health care quality while the present quantitative data mining study looked into the relationship pattern between the cost of living and health care quality among selected Asian cities. The utilized data for cost of living and health care quality indices were taken from Numbeo. The present study also examined the health care system, health care expenditure, private insurance coverage and government subsidy, health industry employment and health care education among the 36 Asian cities.

The study was conducted to provide baseline information on the relationship between the cost of living and health care quality. Additionally, it offered pattern or model to describe the characteristics of the two indices. Also, this study ensured that the underlying core components of health be guaranteed to all individuals. Moreover, the study will provide policy-makers, both government, and non-government organizations information which serves as the basis for decision-making relative to health care in the improvement of its services, infrastructure, facilities, government subsidy on health insurances, availability, the increased accessibility, acceptability, and quality. Lastly, the study will help in all phases of a health care program: assessment, analysis, planning, implementation, monitoring and evaluation to ensure optimum quality of health and life for all.

## Objectives of the Study

This study aims to determine the relationships between the cost of living and health care index. Specifically, it sought to:

1. Generate a pattern or model to describe the characteristics of the cost of living and health care index and;

2. Determine the factors in the increase and decrease of the cost of living to health care index.

## Conceptual Framework

The study's framework is anchored from the study of Begashaw and Tesfaye (2015) which showed the quality of health care contained within the cost of living. The concentric circles around the interactions revolve between the four factors, namely:

1. Socio-economic and demographic factors (age, marital status, education,

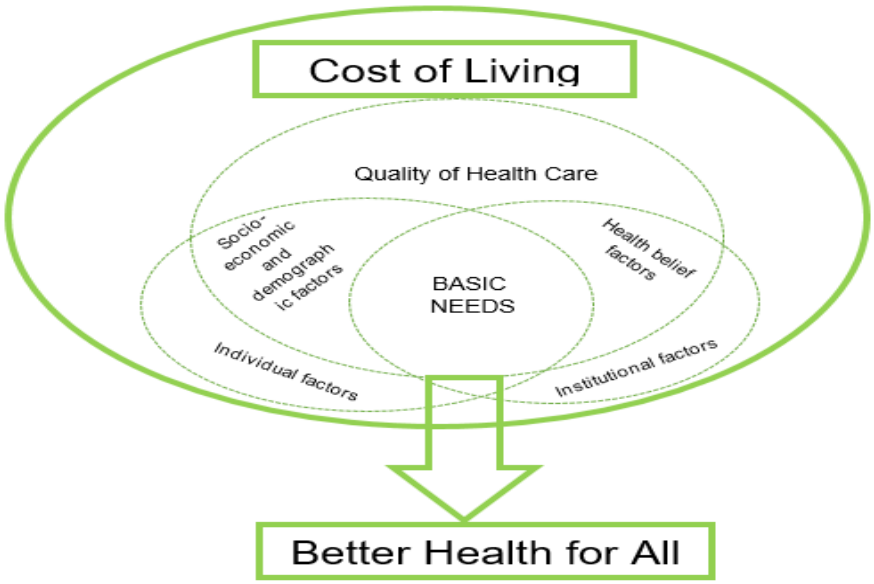


**Figure 1.** *Conceptual framework of the cost of living and the quality of health care*

**Table 1.** Cost of living index versus health care index among Asian cities

| Point | City | X_Cost of Living Index | Y_Healthcare Index |
|---|---|---|---|
| 1 | Bursa, Turkey | 43.79 | 89.08 |
| 2 | Mangalore, India | 21.99 | 77.93 |
| 3 | Riyadh, Saudi Arabia | 47.78 | 70.01 |
| 4 | Taipei, Taiwan | 60.71 | 84.3 |
| 5 | Tokyo, Japan | 102.05 | 78.09 |
| 6 | Abu Dhabi, United Arab Emirates | 60.75 | 54.63 |
| 7 | Tel Aviv-yafo, Israel | 79.19 | 76.01 |
| 8 | Dubai, United Arab Emirates | 69.12 | 61.55 |
| 9 | Pune, India | 25.87 | 63.91 |
| 10 | Bangalore, India | 25.69 | 66.34 |
| 11 | Doha, Qatar | 67.9 | 68.29 |
| 12 | Hyderabad, India | 23.54 | 73.88 |
| 13 | Ankara, Turkey | 45.28 | 67.69 |
| 14 | Istanbul, Turkey | 49.99 | 70.03 |
| 15 | Tbilisi, Georgia | 32.77 | 54.55 |
| 16 | Amman, Jordan | 57.54 | 70.76 |
| 17 | Yerevan, Armenia | 34.25 | 52.78 |
| 18 | Hong Kong, Hong Kong | 78.2 | 75.3 |
| 19 | Beirut, Lebanon | 62.65 | 67.4 |
| 20 | Shanghai, China | 54.36 | 57.54 |
| 21 | Baku, Azerbaijan | 31.22 | 35.88 |
| 22 | Singapore, Singapore | 87.83 | 70.86 |
| 23 | Tehran, Iran | 41.88 | 49.71 |
| 24 | Chennai, India | 25.43 | 74.98 |
| 25 | Delhi, India | 27.63 | 64.68 |
| 26 | Beijing, China | 51.64 | 63.64 |
| 27 | Colombo, Sri Lanka | 35.99 | 80.34 |
| 28 | Karachi, Pakistan | 28.44 | 55.5 |
| 29 | Davao, Philippines | 32.5 | 71.9 |
| 30 | Jakarta, Indonesia | 45.42 | 63.9 |
| 31 | Bangkok, Thailand | 49.83 | 80.72 |
| 32 | Kolkata, India | 25.22 | 52.73 |
| 33 | Mumbai, India | 28.14 | 59.56 |
| 34 | Manila, Philippines | 40.82 | 68.29 |
| 35 | Kuala Lumpur, Malaysia | 44.51 | 60.7 |
| 36 | Ho Chi Minh City, Vietnam | 41.74 | 48.4 |

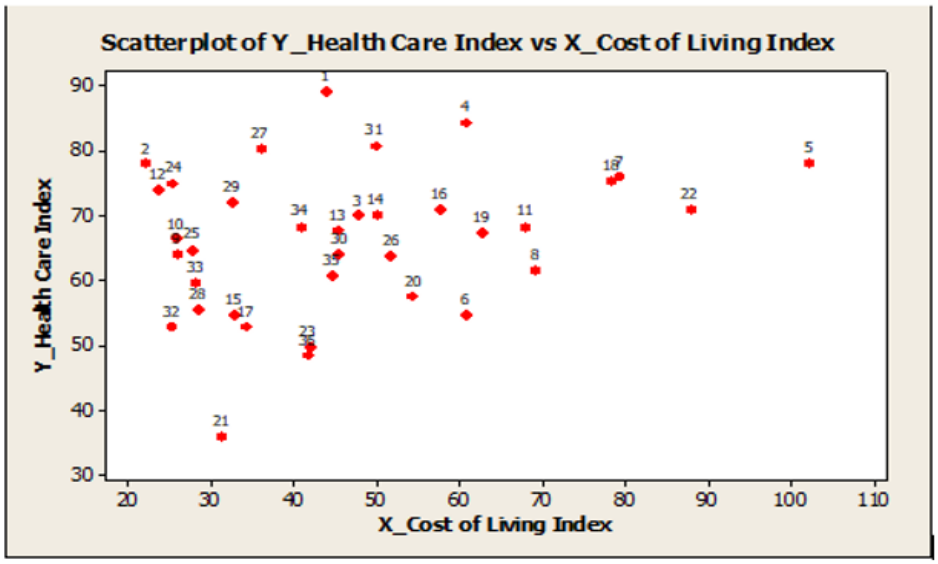


**Figure 2.** *A plot of Health Care Index versus Cost of Living Index*

occupation, family size and family income);

2. Individual factors (self-medication or traditional medicine);

3. Institutional factors (price of health service and distance from health facility) and;

4) Health belief factors (attitude towards health services) illustrating that the quality of health care is vital in the realization of basic human needs.

The arrow shows that the cost of living, regardless of many factors, contributes to the broader aim of a better health for all.

The cost of living comprises the basic human needs as food, clothing, housing, work and education among others. The dashed lines blur the boundaries of the four factors and the quality of health care under the cost of living. Within the space of interaction between the different aspects, the framework shows the level of correlative relationship. At the most basic level, better health for all requires professional and educational standards according to the Faculty of Public Health (FPH)

## Methodology

This study used a quantitative method utilizing data mining technique in the data collection to develop a model on the relationship of cost of living and health care indices among 36 cities in Asia.

Statistical treatment was employed to treat data taken from Numbeo in determining the factors of the cost of living to the increase and decrease in the Health Care Index.

Table 1 shows the thirty-six (36) cities in Asia with the cost of living and health care index based on Numbeo (2016) data. These data sets were plotted using a statistical software.

The result was analyzed to identify the relationship between the cost of living and health care indices of the 36 cities in Asia. Different patterns were formulated to uncover the relationships between these two factors.

## Result and Discussions

Different patterns were derived from the scatterplot of the cost of living (X) and health care index (Y). Points which are out of the outline and which are considered nuisance were removed. One of the patterns observed was a simple linear model which has an

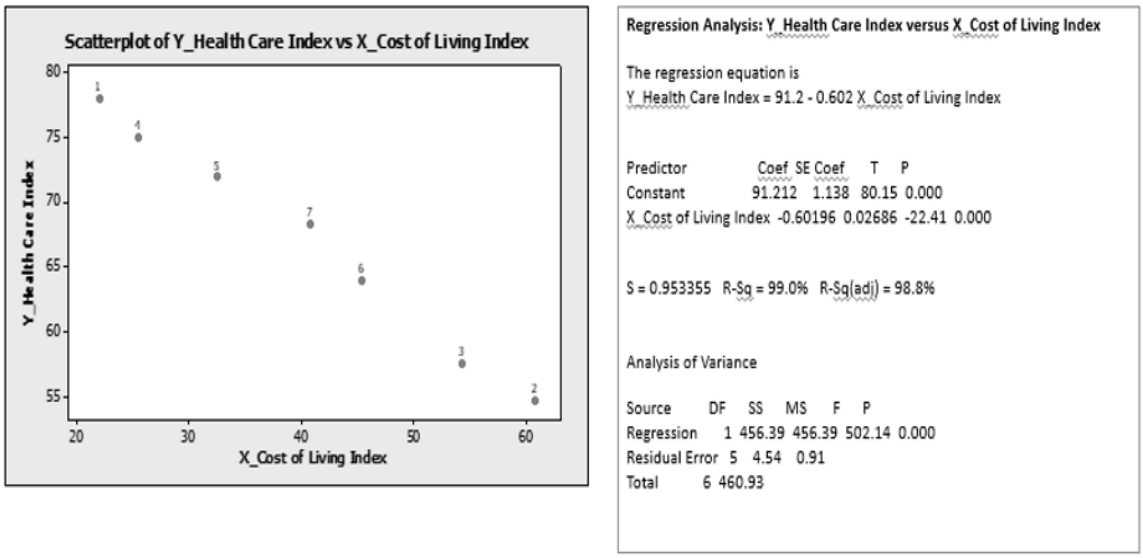


**Figure 3.** *Simple linear model (inverse relation) with regression analysis of Health Care Index versus Cost of Living Index*

**Table 2.** Indices of cost of living and health care among Asian countries in a simple linear model

| City | Gross Domestic Product | Tax rate |
|---|---|---|
| Mangalore, India | 2288.72 | 34.61 |
| Abu Dhabi, United Arab Emirates | 325.135 | 55 |
| Shanghai, China | 11383.312 | 25 |
| Chennai, India | 11383.312 | 34.61 |
| Davao, Philippines | 310.312 | 30 |
| Jakarta, Indonesia | 936.955 | 25 |
| Manila, Philippines | 310.312 | 30 |

inverse relationship between the cost of living and health care Index.

The graph shows a Simple Linear Model (Inverse Relation) with one predictor X= Cost of Living Index and a response variable Y= Health Care Index. A regression analysis was computed using the regression equation $Y = 1 + 1.5\ X - 2.5X\hat{2}$ with R-Sq (adj) = 98.8%. In Asian cities, the relationship between the cost of living and Health Care Index varies from one city to another. Around 19% of these are composed of cities where the people have a high cost of living and can only avail of a few health care services. These cities were Mangalore, Abu Dhabi, Shanghai, Chennai, Davao, Jakarta, and Manila as shown in the table 2.

Cities such as Chennia, Mangalore, Davao, Manila, and Jakarta have a high quality of health care system with more than 60% while cities such as Abu Dhabi, Shanghai and Jakarta have a high cost of living with more than 40%. In countries where the quality of health care system is high, people did not rely on government health care programs and services. They tend to allocate a great amount of their income for health care. In countries like UAE and China, the cost of living is high, and

Regression Analysis: Y_Health Care Index versus X_Cost of Living Index

The regression equation is
Y_Health Care Index = 18.2 + 0.599 X_Cost of Living Index

| Predictor | Coef | SE Coef | T | P |
|---|---|---|---|---|
| Constant | 18.186 | 1.996 | 9.11 | 0.003 |
| X_Cost of Living Index | 0.59857 | 0.02687 | 22.27 | 0.000 |

S = 1.45598 R-Sq = 99.4% R-Sq(adj) = 99.2%

Analysis of Variance

| Source | DF | SS | MS | F | P |
|---|---|---|---|---|---|
| Regression | 1 | 1051.6 | 1051.6 | 496.08 | 0.000 |
| Residual Error | 3 | 6.4 | 2.1 | | |
| Total | 4 | 1058.0 | | | |

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**Figure 4.** *Simple linear model (directly proportional) with regression analysis of Health care index versus Cost of Living Index*

**Table 3.** Indices of cost of living and health care among Asian countries in a simple linear model

| Rank | City | X_Cost of Living Index | Y_Healthcare Index |
|---|---|---|---|
| 5 | Tokyo, Japan | 102.05 | 87.07 |
| 6 | Abu Dhabi, United Arab Emirates | 60.75 | 57.71 |
| 8 | Dubai, United Arab Emirates | 69.12 | 61.55 |
| 21 | Baku, Azerbaijan | 31.22 | 35.88 |
| 22 | Singapore, Singapore | 87.83 | 69.98 |

the government health expenditure budget is also high. Therefore, people are not particular of the quality of health care. They are just dependent on the health services offered by the government.

Another pattern was a simple linear model which is directly proportional to the cost of living and Health Care Index. The graph shows a Simple Linear Model (Directly Proportional) with one predictor X= Cost of Living and a response Variable Y= Health Care Index. A regression analysis was computed using the regression Y= 1 + 1.5 X – 2.5 X$\hat{2}$ with R-Sq (adj) = 99.2%. In Asian cities, the relationship between the cost of living and Health Care Index differ from one city to another. Around 14% of Asian cities have people with the low cost of living who still avail of health care services.

Cities such as Tokyo, Japan, and Singapore have the highest cost of living with more than 80% while cities such as Dubai, and Abu Dhabi follows with more than 60%. Among the five cities mentioned, Baku has the lowest cost of living which is less than 40%. In a highly developed country like Japan where the quality of health care delivery system is high, the government is paying 70% for all of the health care delivery system, and the patient is pays only 30%. Also, all residents are required by the government to have health insurance coverage.In Abu Dhabi and Dubai,

both cities from the United Arab Emirates, the cost of living is also high, but they have highly developed health services. That is why health insurance in these countries is not compulsory. The people of Baku, Azerbaijan are not particular with the health care delivery system.

An elliptical pattern was observed as shown in Figure 4 which depicts the relationship between the cost of living and health care indices among the 36 cities in Asia. Nuisance Points 1, 2, 32, 21, and 5 were taken out from the dataset. An elliptical pattern was observed with the center point at (56.76, 66.35) and with the variance of (62.14, 35.9) along the cost of living index(x) and the Health Care Index(y).

Cities that surround the point of reference are Bursa Turkey, Beijing China, Amman Jordan, Shanghai China, and Beirut Lebanon. The living expenses in these Asian cities are significantly higher in comparison to other cities. The cost of housing, food, and energy largely account for the high cost of living. In Amman Jordan, the high cost of living problem is not related to the increased outlays of commodities. However, salaries are not high, make health care more affordable.

The quality of health care, on the other hand, varies from cities to cities. In Bursa Turkey, health care is cheaper, and there are many private and public hospitals across the country where people have access to good quality private hospitals with experienced doctors and medical staff that most of which can speak English. Health care in China, Lebanon and Jordan is a significant point of contention. The Chinese health care system is hospital-centered and most of residents take out private health insurance and seek treatment at private facilities. Jordanians are also accustomed to receiving a high standard of medical care with the expertise of Jordanian doctors. In Beirut Lebanon, there is a wide range of both private and public health care with a high standard medical and dental care.

From the variance of the cost of living (x)= 62.14, Abu Dhabi is one of the cities that is highly variable regarding the cost of living. An individual's cost of living also is very much unpredictable. With regards to their health care system, both public and private health care services are available. Since their health care is accompanied by high prices, their health insurance is not compulsory. On the other hand, they are being encouraged to get health insurance which covers most of the costs. Abu Dhabi has a high cost of living where apply few people can avail of their health care services.

On the other hand, Taipei, Taiwan has a much lower cost of living. As to their health care delivery system, they have an unbelievably low-priced health insurance or Medicare. Majority of the citizens make use of the government funded health care through their National Health Insurance. It enables the members to access medical benefits like emergency care, doctors and dental consultations, maternity care and even access to Traditional Chinese Medicine. Through the National Health Insurance, the citizens can easily access all medical benefits. Since the living expenses are low and the health care delivery system is very affordable, there is a possibility that many residents will invest more for their health.

The variance of the Health Care Index(y)= 35.9 indicates that cities such as Bangalore India and Singapore have a similar cost of living but the health care varies. Indias considered as one of the cheapest cities in the world have cities Bangalore and Chennai, which reflect the structurally low prices enjoyed on the subcontinent. Although India is prospected for future growth, much of this is driven by its large population and the untapped potential within the economy. Income inequality means that low wages multiply, driving down household spending and creating many tiers of pricing that keep spending per head low. Low-cost and abundant supply of commodities into cities from rural producers with short supply chains, as well as government subsidies on some products, has kept prices down (WCOL, 2015). The cost of living is practically low in

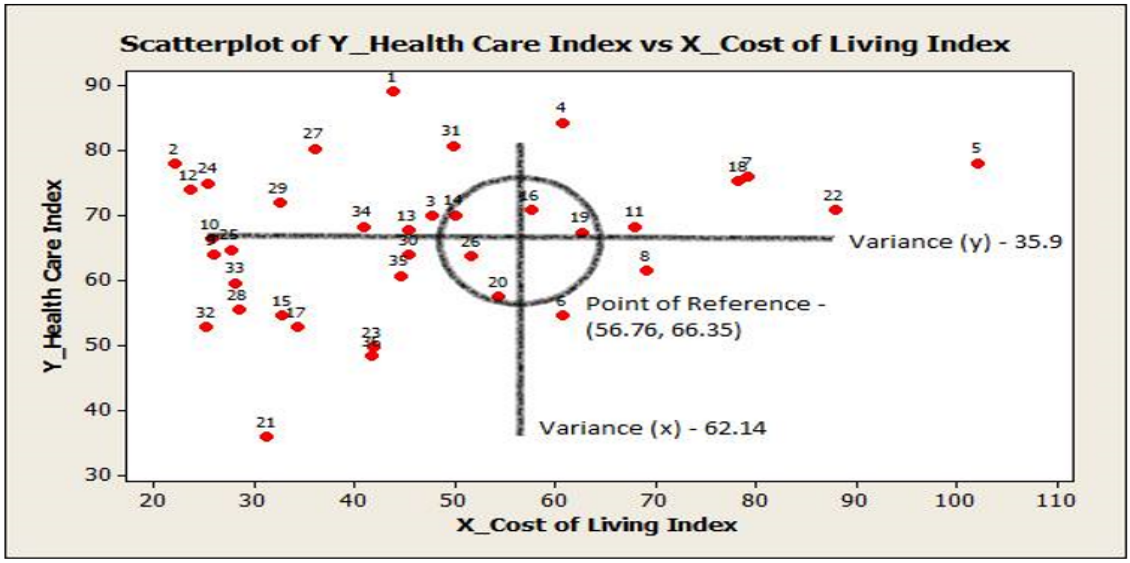


**Figure 5.** *An elliptical pattern of the Cost of Living Index versus Health Care Index*

India. The standard of living in India shows large disparity from state to state.

On the other hand, Singapore is one of the best countries to live. Its health care deems to be of high standard. Amidst the soaring cost of living in Singapore, the Health care Index is high since the government put in place a planned, built, developed and maintained public health care system. The public and private system are administered by the Ministry of Health which takes full responsibility for assessing health needs and planning and delivering services through networks of health and hospital facilities, day care centers, and nursing homes. Universal health care coverage is afforded by Singaporean government to all citizens with well-developed financing system. Funds for the Health care System emanated from general revenue and used for subsidies, campaigns for health practices promotion, development, training, and infrastructure expenses. Private insurance supplements the provision afforded by the government. Moreover, the employers may offer private insurance to their employees as a staff benefit. Typically, employer-sponsored insurance covers the primary care and other outpatient visits, in addition to hospitalizations (Ministry of Health, 2013).

## Conclusions and Recommendations

This study aimed at determining the relationships between the cost of living and health care indices. Firstly, it was designed to generate a pattern or model to describe the characteristics of the cost of living and Health care Index. There were three patterns observed from the scatterplot, namely 1) simple linear model (inverse relation), 2) simple linear model (directly proportional) and 3) elliptical pattern. In general, people living in Asia does not necessarily avail of health care programs and services regardless of their cost of living. Such relationship, therefore, differs from one country to another. Nevertheless, living index and Health care Index relationships infers.

Secondly, this study aimed to determine the factors in the increase and decrease of the cost of living to Health Care Index. In summary, there are cities in Asia that tend to increase its Health Care Index when the cost of living increases. These people are health conscious and allocate a higher percentage of their expenses to health and

wellness. These cities have minimal budget allocation for health care from their GDP which requires their people to apportion more personal funds for health. Thus, to solve the increased cost of living, the budget allocation for health care becomes dynamic. Some cities in Asia decreased its Health Care Index when the cost of living increased. Despite higher budget allocation for health care by the government from their GDP, the cost of living is still relatively higher wherein more of people's expenses are appropriated on other personal needs rather than health services. Therefore, government budget for health care remains static despite the increased cost of living among its people.

For future studies, the researchers recommend studies on quality improvement which looks on the overuse, underuse or misuse of health care services, which include the following:

- Identify factors which encourage quality and various payment facility scheme, financial incentives, and government factors which affect the behavior of health care organizations;
- Design and implement new health care policies, processes and unifying framework to provide safer, high-quality care; and,
- Mainstream human rights, gender equality, and equity into health care programs.

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